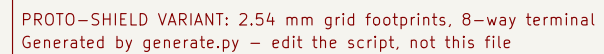


E-stop loops: 3.3 k  $\rightarrow$  6.3 mA (0.20 W at 28.8 V). Lidar inputs: 4.7 k  $\rightarrow$  4.4 mA (0.14 W). R1-R4 are 1/2 W (body 9 x 3.2 mm); pull-ups 1/4 W. No current (button pressed, zone occupied, wire cut, 24 V lost)  $\rightarrow$  pin HIGH = STOP. F1 protects the 24 V wires going out to the buttons. GND\_24V and GND are NOT connected: this is the isolation barrier. Pull-ups go to IOREF (5 V on UNO R4 WiFi, 3.3 V on UNO Q), never to 5V. C1-C4 are OPTIONAL: fitted, an opening shorter than about 1.5 ms is not seen by the Arduino, and every stop is seen about 1.6 ms later. E-stop 1: button released (contact closed)  $\rightarrow$  D2 LOW, LED lit



**Title: Safety inputs: 2 e-stops + 2 lidars**

|                     |                  |          |
|---------------------|------------------|----------|
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